



Tubular-aware mamba for accurate retinal vessel segmentation: preserving fine details and topological connectivity

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Abstract

Accurate retinal vessel segmentation is crucial for diagnosing ocular and systemic diseases. Current methods struggle to balance noise discrimination with detail preservation, especially for capillary detection. This paper introduces TA-Mamba, a novel framework combining the State Space Model (SSM) Mamba with tubular structure perception modules. TA-Mamba utilizes a high and low frequency attention Mamba block, tubular-aware gated convolution, serpentine spatial convolution, directional feature fusion, and a convolution-up block. Experimental results on DRIVE, CHASE_DB1, and STARE datasets demonstrate TA-Mamba's superior performance, outperforming state-of-the-art methods in accuracy, continuity, and topological connectivity. This work represents a significant advancement in retinal vessel segmentation, effectively addressing challenges posed by low contrast and high noise in retinal images.

Keywords Retinal vessel segmentation · Capillary detection · State space model · Mamba

1 Introduction

For downstream procedures to be accurate and efficient, retinal vessel segmentation must be done precisely. A good picture of vessel architecture helps doctors diagnose illnesses in clinical situations. However, manual segmentation takes much time and demands a high level of knowledge. As a result, precise and thorough retinal vessel segmentation is essential. This procedure has several difficulties: (1) The global morphology is complicated and varied, and the images are characterized by low contrast and high noise levels, which hinder the model's generalization capabilities; (2)

The local capillaries are fragile, making it challenging for models to detect minor changes.

In recent years, many deep-learning-based methods for medical image segmentation have been proposed, and have shown promising results. Most of them follow the encoder-decoder architecture similar to U-Net [28]. However, we find this architecture challenging to precisely distinguish targets from noise in our task. Enhancing the model's capability to discern capillaries of diverse morphologies and low contrast is essential for improving segmentation performance. Sabina et al. [21] proposed an effective algorithm for detecting nasolabial wrinkle lines based on the Active Appearance Model (AAM) and Hessian filter. This method is capable of tracking curvilinear structures, thereby identifying complexly structured wrinkles. This presents a viable method for our task, capable of identifying complexly structured curves by monitoring curvilinear structures. However, this approach has not been employed in our study. Dai et al. [22] developed a deformable convolution that enhances the network's adaptability to the target object and boosts localization accuracy by dynamically adjusting the convolution kernel's geometry based on the input feature map's geometric variations. However, capturing global relationships remains a challenge for CNN-based methods due to the inherently local nature of convolutional layers [23]. Multi-scale feature fusion has a wide range of applications in the

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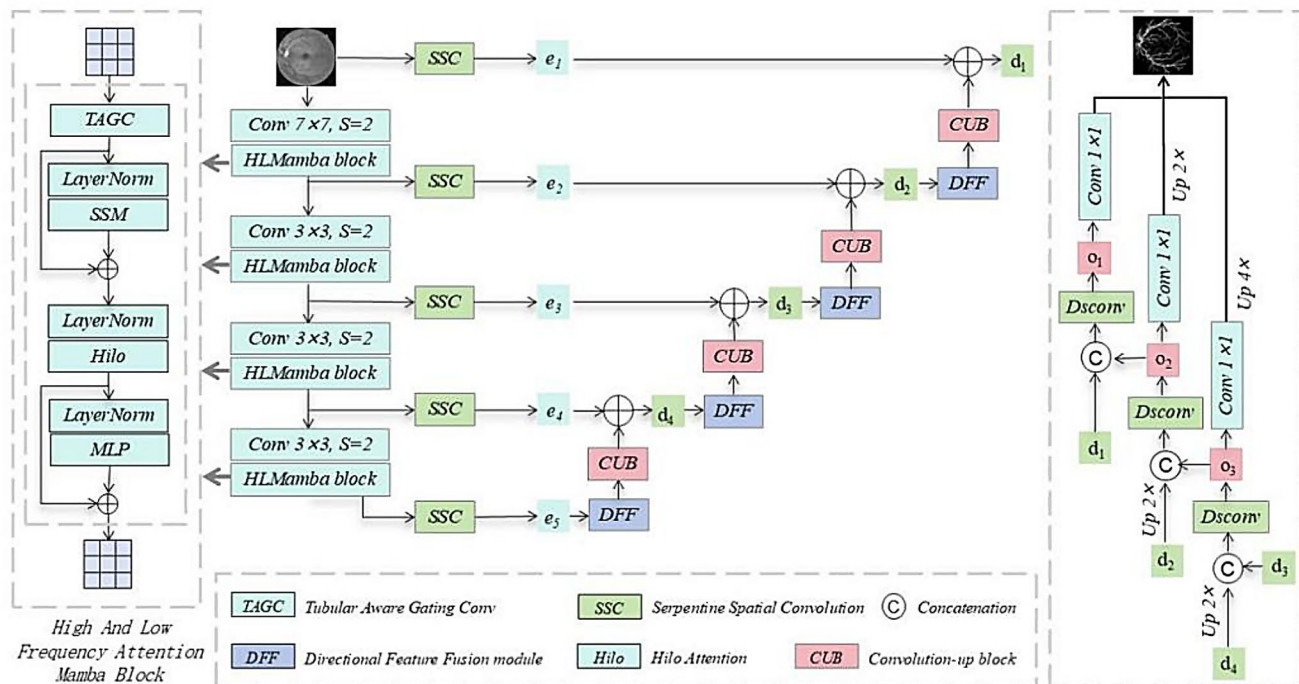


Fig. 1 The overview of the proposed TA-Mamba

field of image segmentation [25, 26], which improves the performance of the model by integrating features at different scales, enabling it to better understand and process complex visual information.

Transformer architectures [24, 29] can extract global information, but it incurs a significant computational burden when dealing with overly long feature sequences. To overcome the limitations of the Transformer architecture in modeling long sequences, Mamba1 [15], as a structured state space sequence model (S4) with an integrated selection mechanism, reduces the computational complexity of the model and demonstrates enhanced capabilities in handling long-range dependencies. Mamba2 [18] can process long sequences and excels in tasks requiring fine-grained contextual understanding. The U-Mamba algorithm [16] introduces a new hybrid framework that integrates State Space Models (SSM) with Convolutional Neural Networks (CNN), marking its initial venture into the domain of medical image segmentation. Furthermore, SegMamba [17] implements an integrated approach by incorporating SSM within the encoder component and utilizing CNN for the decoder, thereby crafting a hybrid SSM-CNN model specifically designed for the segmentation of 3D brain tumors. Despite these advancements, the efficacy of SSM-driven models in segmenting medical images with complex topological structures, like those found in retinal vessels, remains largely unexplored.

Inspired by the above, our work introduces an innovative framework termed TA-Mamba. This framework integrates Mamba with a tubular structure perception module. To the best of our knowledge, this is the first method specifically utilizing Mamba for segmenting medical images with tubular topological structures. Our specific contributions are as follows:

1. A high and low frequency attention Mamba block, captures global contextual information by focusing on different frequency bands.
2. A tubular-aware gated convolution intensifies the feature representation of the spatial dimensions preceding each Mamba module, thereby boosting the model's ability to capture fine and thin features.
3. A serpentine spatial convolution for further extraction of tubular features.
4. A directional feature fusion module that isolates the directional features of the target, refining the model's perception of directional features for efficient feature integration.
5. A convolution-up block that optimizes computational efficiency without sacrificing performance, enhancing the upsampling process during segmentation.

High and low frequency attention Mamba block helps us capture global contextual information to distinguish targets from noise precisely. Tubular-aware gated convolution,

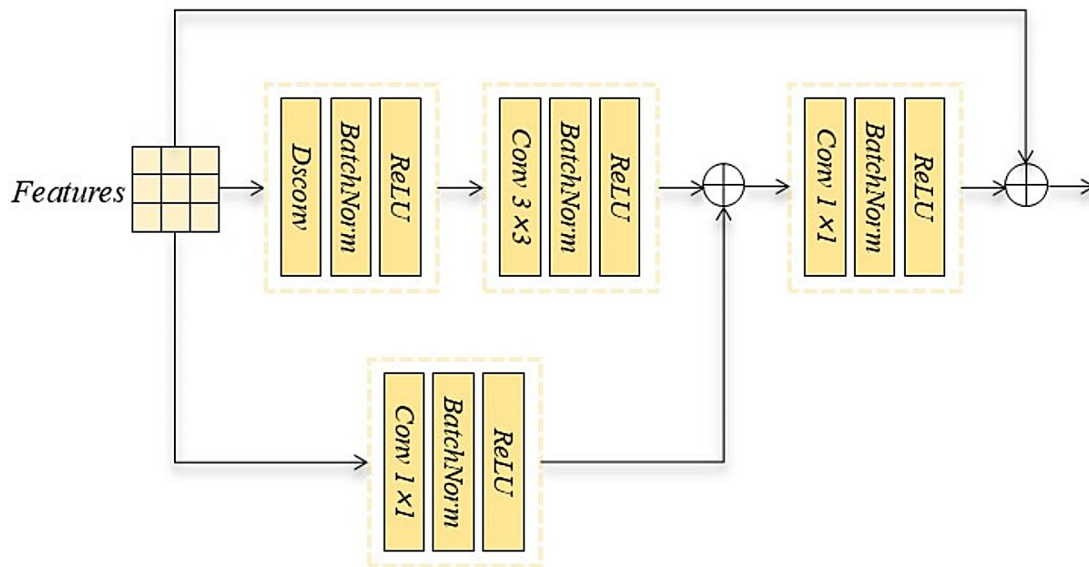


Fig. 2 The tubular-aware gated convolution

serpentine spatial convolution, and directional feature fusion module assist us in capturing capillaries. Our framework effectively addresses the challenges in retinal vessel segmentation.

Experimental results on the DRIVE, CHASE_DB1, and STARE datasets demonstrate that our method provides improved accuracy and continuity in the segmentation of capillaries. These contributions jointly advance the state-of-the-art in retinal vessel segmentation.

2 Method

In this section, the network structure of TA-Mamba is first introduced, followed by a detailed description of high and low frequency attention Mamba block, tubular-aware gating convolution, directional feature fusion module, convolution-up block, and serpentine spatial convolution.

2.1 Network structure

The specific architecture of TA-Mamba is depicted in Fig. 1 and primarily consists of four components: (1) a feature encoder with high and low frequency attention Mamba (HLMamba) block (2) a serpentine spatial convolution (SSC), (3) a decoder composed of a directional feature fusion (DFF) module and a convolution-up block (CUB), and (4) a decoder based on Dynamic Snake Convolution (DSConv) [6].

This study introduces a HLMamba block following the convolutional layers of the feature encoding block. The feature encoder extracts salient features from the input images

and progressively reduces the spatial dimensions of the input data, consisting mainly of four layers, each incorporating a convolutional layer with a stride of 2 and a HLMamba block. The original input and the output of each layer are separately processed through SSC to further extract tubular features. A DFF module is then added to reinforce the directional information of the output features, and a CUB performs the upsampling of the features. The semantic aggregation module integrates low-level spatial details with high-level abstract semantic information, enhancing the network's comprehension and processing capabilities. The feature decoder also follows a four-layer structure, with each layer including a DFF module, a CUB, and a semantic aggregation module. Finally, to further predict the segmentation results, the upsampled features from the first layer of the decoder are concatenated with the features from the second layer of the decoder. Subsequently, the connected features are processed by DSConv. Thereafter, the processed features are subjected to the aforementioned operations with the decoder's features. Each output of the DSConv is then fed into a 1×1 convolution. Ultimately, we integrate three distinct prediction maps to generate the final segmentation output.

2.2 High and low frequency attention Mamba (HLMamba) block

HLMamba block is the core component of TA-Mamba architecture, as illustrated on the left side of Fig. 1. In the case of HLMamba block, the feature initially undergoes a TAGC, followed by layer normalization. Subsequently, the feature is input into a state space model (SSM) [18] for

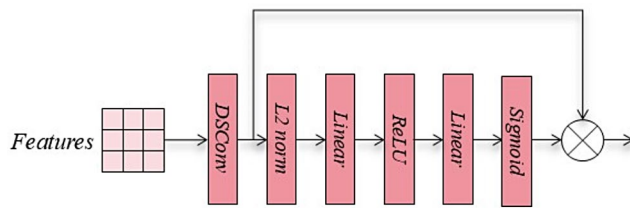


Fig. 3 The serpentine spatial convolution

further feature extraction. Thereafter, layer normalization is applied to the feature, which is then subjected to Hilo Attention [19] to capture both global dependencies and fine-grained local relationships within the feature. Finally, the feature is enriched through layer normalization and a multilayer perceptron (MLP), with two skip connections interspersed to facilitate feature fusion. The computational process can be defined as follows:

$$\begin{aligned} \hat{f} &= TAGC(f) & \hat{f} &= Mamba(LN(\hat{f})) + \hat{f} \\ \hat{f} &= Hilo(LN(\hat{f})) & \hat{f} &= MLP(BN(\hat{f})) + \hat{f} \end{aligned} \quad (1)$$

Tubular-aware gated convolution (TAGC). The Mamba layer represents interdependencies between features through the transformation of two-dimensional feature maps into a linear one-dimensional array. To address the need for spatial relationship recognition at stages preceding the Mamba layer, we design a TAGC. Through experiments, we found that it can extract tubular structures with greater accuracy than ordinary gated convolution. As depicted in Fig. 2, the input features are conveyed through two successive blocks (the first dynamic snake convolutional block comprises a dynamic snake convolution, a normalization layer, and

a nonlinear layer, while the second convolutional block includes a regular 3×3 convolution, a normalization layer, and a nonlinear layer) and a convolutional block (this block contains a 1×1 convolution, a normalization layer, and a nonlinear layer), yielding two novel features. These features are then aggregated pixel-wise, emulating a gating mechanism to regulate information transfer. Subsequently, a convolutional block is utilized to further integrate the features, resulting in the reuse of input features through a residual connection.

$$TAGC(f) = C^{1 \times 1}(C^{3 \times 3}(DSC(f)) + C^{1 \times 1}(f)) + f \quad (2)$$

Where the f denotes the input 2D features, the DSC represents the dynamic snake convolutional block, and the C signifies the convolutional block.

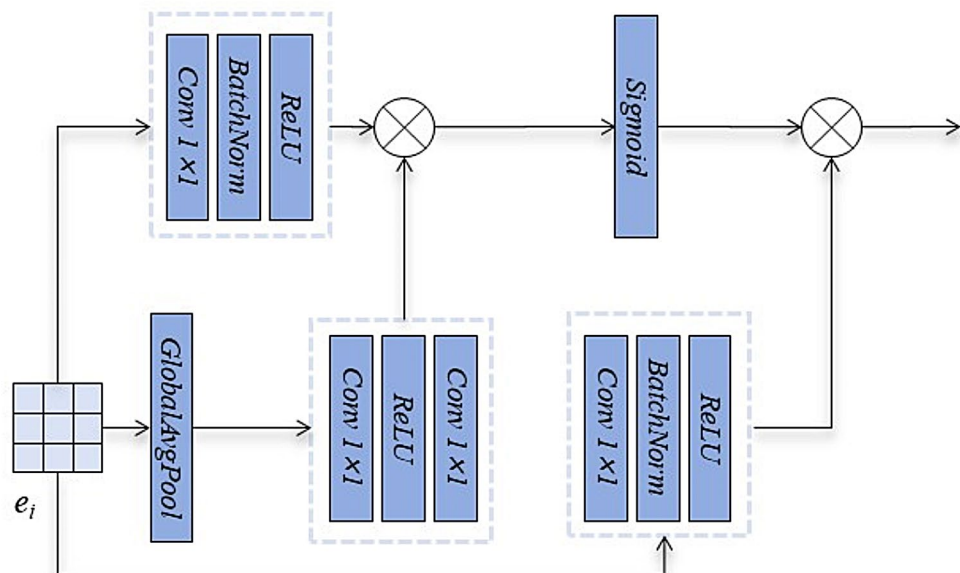
2.3 Serpentine spatial convolution (SSC)

DSConv has been proven superior in the task of segmenting retinal vessels [6]. Inspired by this, we design a SSC. The structure of the SSC is illustrated in Fig. 3, where the feature x is first processed through DSConv to yield the feature x_s .

$$x_s = DSConv(x) \quad (3)$$

The L2 norm of x_s is then calculated, followed by its progression through two fully connected layers (the first comprising a linear transformation and a non-linear activation function ReLU, and the second consisting of a linear transformation and a non-linear activation function Sigmoid) to produce the weight tensor y .

Fig. 4 The directional feature fusion Module



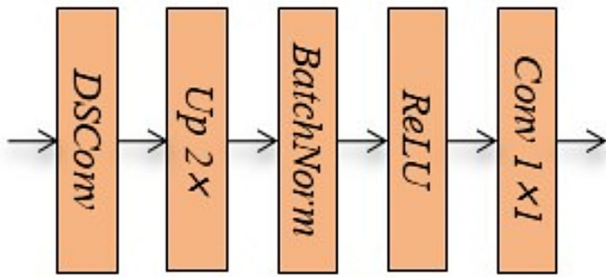


Fig. 5 The convolution-up block

$$y = FC2(FC1(\|x_s\|_2)) \quad (4)$$

Finally, x_s is element-wise multiplied with y to obtain e_i , where i denotes the i^{th} SSC. Notably, in this scenario, feature aggregation based on the norm, specifically the L2 norm, results in enhanced performance compared to global average pooling [14].

$$e_i = x_s y \quad (5)$$

2.4 Directional feature fusion (DFF) module

In the segmentation of retinal vessels, it is crucial to maintain the topological relationships and directional information between pixels. Consequently, to augment the network's capacity to capture features of varying orientations within the image, we devise a DFF module, as depicted in Fig. 4. We subject the output e_i of the encoder to a global average pooling followed by a directional feature enhancement block, which includes a 1×1 convolution, a nonlinear layer, and another 1×1 convolution, to extract directional features. Subsequently, a 1×1 convolutional block is utilized to re-encode e_i , and the re-encoded features are multiplied with the directional features pixel by pixel. Finally, a Sigmoid function is employed to ascertain weights for distinct orientations. The directional weights are multiplied with the features of e_i that have been re-encoded through a 1×1 convolutional block pixel by pixel, culminating in the acquisition of directionally enhanced feature x_n , where n denotes the n^{th} DFF module.

$$x_n = \sigma(D^{1 \times 1}(GAP(e_i))C^{1 \times 1}(e_i))C^{1 \times 1}(e_i) \quad (6)$$

Where the D denotes the directional feature enhancement block, and the C represents the convolutional block.

2.5 Convolution-up block (CUB)

To minimize performance loss during upsampling, we design a CUB, as shown in Fig. 5. Experimentally, it performs better than normal upsampling in our task. Each CUB

layer consists of a DSConv, an upsampling with scale-factor 2, a batch normalization, a ReLU activation, and a 1×1 convolution. The CUB can be formulated as the Equation below:

$$d_n = C^{1 \times 1}(U(DSConv(x_n))) \quad (7)$$

Where the $DSConv$ denotes the DSConv, the U represents the upsampling and the $C^{1 \times 1}$ represents the 1×1 convolution.

3 Experiment

In this section, three different databases, the metrics for evaluation, and the experimental setup are introduced.

3.1 Datasets

In the present study, to substantiate the efficacy of the proposed TA-Mamba algorithm, we elect to conduct experimental assessments on three renowned public datasets, namely DRIVE [1], CHASE_DB1 [2], and STARE [3].

The DRIVE dataset comprises 40 color retina images, each with a resolution of 564×584 pixels. By the official partitioning criteria of the dataset, 20 images are allocated for the training phase, while the remaining 20 images are designated for the testing phase.

The STARE dataset includes 20 color retina images, each showing either a normal or a pathological retina, at a resolution of 700×605 pixels. For the purposes of this research, the initial 10 images are utilized as the training set, and the subsequent 10 images as the testing set.

The CHASE_DB1 dataset is constituted by 28 color retina images, with each image having a pixel resolution of 990×960 . In this study, the first 20 images are applied for training purposes and the final 8 images are reserved for testing.

It is noteworthy that the dataset evaluation employed in this manuscript is predicated on the manual segmentation outcomes of the first expert. This assessment methodology affords a precise and dependable benchmark for our algorithm, facilitating the measurement of its performance. By conducting experiments on these widely recognized datasets, we ensure that the assessment outcomes of the proposed method are of a high degree of scientific rigor and comparability.

3.2 Evaluation metrics

We employed Dice, CenterlineDice (clDice) [4], Accuracy (ACC), Area Under CurveROC (AUC), Sensitivity (SE), and Specificity (SP) as the evaluative metrics for our experimental assessments.

3.3 Implementation details

We used PyTorch 2.1.0-cuda11.8 to implement our network and conduct experiments. All of our experiments were conducted on the same device, with the following configuration: CPU: 11th Gen Intel (R) Core (TM) i9-11900 K @ 3.50 GHz, Memory: 128GB, GPU: NVIDIA GeForce GTX 3090 Ti 24GB. For each dataset, we used a batch size of 2. We used combined weighted Cross-Entropy(0.7) and DICE(0.3) loss and an AdamW optimizer [12] with a weight decay of 0.0003 for all experiments. As described in [20], for all datasets, we employed the following preprocessing procedures:

- 1) Conversion of color images to grayscale.
- 2) Normalization of grayscale images to mean and standard deviation.
- 3) Contrast-Limited Adaptive Histogram Equalization(CLAHE) [27].
- 4) Local gamma correction.

DRIVE dataset. We employed random cropping to dimensions of 224×224 pixels and used a warm-up cosine annealing schedule to train the DRIVE dataset, with the maximum learning rate set to 0.0008, the minimum learning rate set to 0.0001, the warm-up period configured for 400 epochs, and the total training duration spanning 1000 epochs.

CHASE_DB1 dataset. We set the random cropping size to 336×336 pixels and used an AdamW optimizer with a learning rate of 0.0005.

STARE dataset. We resized the image to 224×224 pixels and used a warm-up cosine annealing schedule, with the maximum learning rate set to 0.0007, the minimum learning rate set to 0.0001, the warm-up period configured for 400 epochs, and the total training duration spanning 1000 epochs.

4 Results

In this section, we designed experiments to compare with state-of-the-art algorithms and conducted ablation studies on all components to extensively evaluate the proposed method.

4.1 Quantitative evaluation

Our evaluation of TA-Mamba against a set of seven sophisticated segmentation techniques demonstrated that our method held its own in terms of performance.

DRIVE dataset. As can be observed from Table 1, our TA-Mamba outperformed the other six state-of-the-art methods across every metric, with a Dice of 82.48%, a cIDice of 83.21%, an ACC of 96.99%, an AUC of 98.64%, an IOU of 70.21%, a SE of 84.76%, and a SP of 98.82%. Notably, the cIDice surpassed the best-performing method by 1.14%, and the SE exceeded the best-performing method by 1.76%. This suggests that our method maintained superior topological connectivity during segmentation and offered enhanced precision.

CHASE_DB1 dataset. In Table 2, the results obtained by our proposed TA-Mamba are as follows: Dice at 81.68%, cIDice at 84.48%, ACC at 97.69%, AUC at 98.99%, IOU at 69.08%, SE at 87.28%, and SP at 99.01%. In the context of retinal vessel segmentation, cIDice is considered more critical than the traditional Dice coefficient. Although DconnNet [7] exhibited a 0.12% higher Dice score and a 0.32% higher IOU score than ours, our cIDice score surpassed DconnNet by 1.18%. Collectively, our method demonstrates superior performance compared to DconnNet. Although FR-UNet [13] achieved a better 0.14% AUC score than our TA-Mamba, our approach outperformed FR-UNet across all other metrics. We can conclude that our TA-Mamba also performs commendably on the CHASE_DB1 dataset.

STARE dataset. Due to the inconsistency in the partitioning of the training and validation sets for the STARE dataset among various methods, a direct comparison would be inequitable. Therefore, we replicated four advanced methods, utilizing the first ten images of the STARE dataset as the training set and the subsequent ten images as the validation set, with the results present in Table 3. The TA-Mamba exhibited a Dice of 81.59%, a cIDice of 85.15%, an ACC of 97.38%, an AUC of 98.67%, an IOU of 69.12%, a SE of 82.73%, and an SP of 99.47%. Although DSCNet [6] surpassed us by 0.16% in the SP score, TA-Mamba demonstrated superior performance across all other metrics in comparison to DSCNet. This indicates that TA-Mamba also performs well on the STARE dataset.

Table 1 Experiment results on the DRIVE dataset. Bolded data indicate the best. This kind of display continues for all subsequent experiments

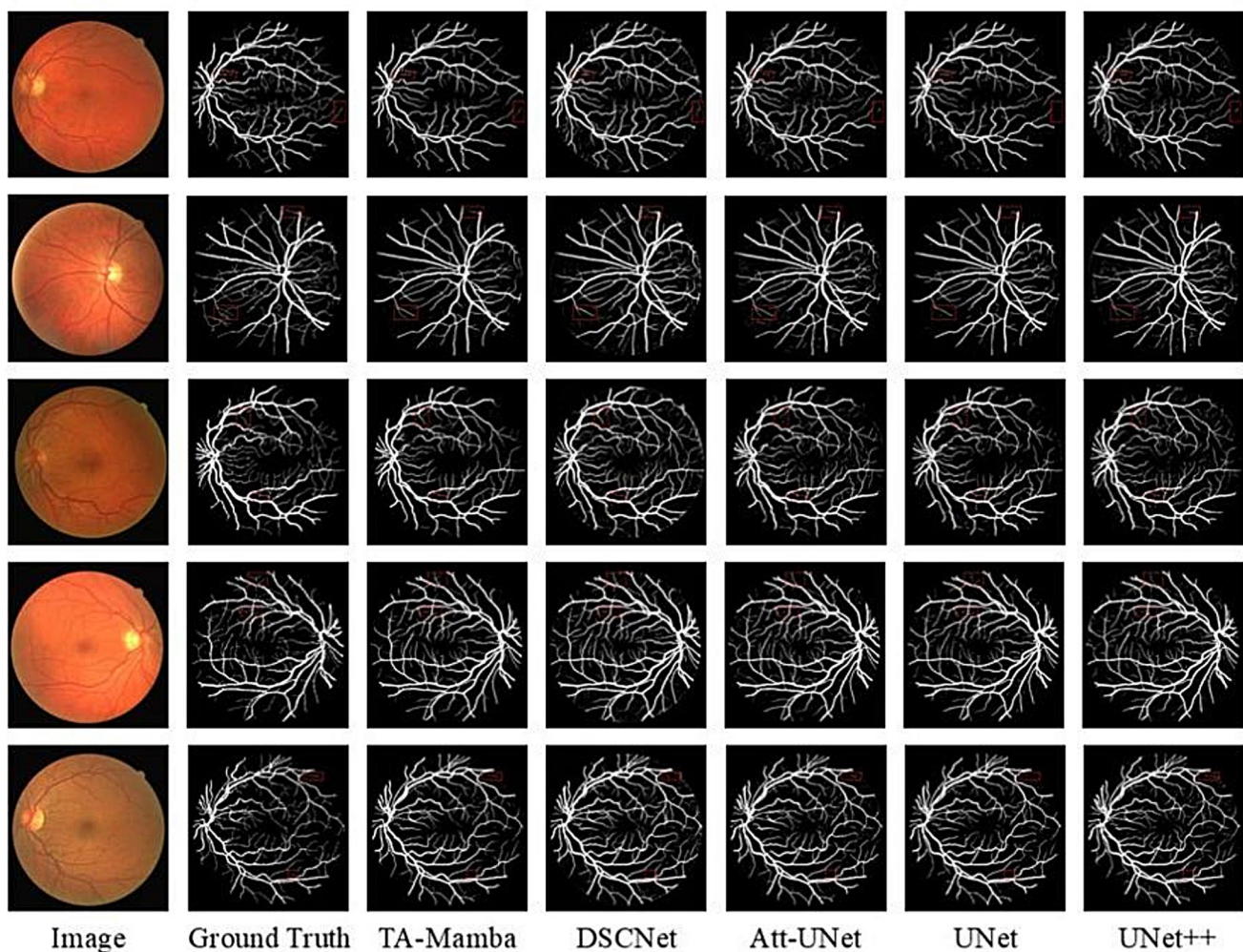
Methods	Dice	cIDice	ACC	AUC	IOU	SE	SP
UNet [5]	81.41	79.66	96.78	98.25	68.64	80.57	98.33
UNet++ [10]	81.14	81.08	96.79	98.25	68.27	78.91	98.50
Att-UNet [11]	80.39	81.55	96.62	97.74	67.21	79.06	98.31
DSCNet [6]	82.06	82.07	96.87	90.27	-	-	-
SCS-Net [8]	-	-	96.97	98.37	-	82.89	98.38
PVT-GCASCADE [9]	82.10	-	96.89	-	69.70	83.00	98.22
TA-Mamba(ours)	82.48	83.21	96.99	98.64	70.21	84.76	98.82

Table 2 Experiment results on the CHASE_DB1 dataset. Bolded data indicate the best. This kind of display continues for all subsequent experiments

Methods	Dice	clDice	ACC	AUC	IOU	SE	SP
UNet [5]	78.98	74.00	97.43	98.36	65.26	76.50	98.84
UNet++ [10]	80.15	83.38	97.39	98.81	66.88	83.57	98.32
Att-UNet [11]	79.64	75.30	97.30	98.48	66.17	83.84	98.20
DconnNet [7]	81.80	83.30	-	-	69.40	-	-
SCS-Net [8]	-	-	97.44	98.67	-	83.65	98.39
FR-UNet [13]	81.51	-	97.48	99.13	68.82	87.98	98.14
TA-Mamba(ours)	81.68	84.48	97.69	98.99	69.08	87.28	99.01

Table 3 Experiment results on the STARE dataset. Bolded data indicate the best. This kind of display continues for all subsequent experiments

Methods	Dice	clDice	ACC	AUC	IOU	SE	SP
UNet [5]	80.66	84.35	97.27	98.37	67.79	82.35	99.43
UNet++ [10]	79.52	83.46	97.16	97.89	66.29	78.54	99.39
Att-UNet [11]	80.55	84.64	97.21	98.40	67.63	80.92	99.20
DSCNet [6]	79.85	82.55	97.16	98.30	66.70	78.88	99.63
TA-Mamba(ours)	81.59	85.15	97.38	98.67	69.12	82.73	99.47

**Fig. 6** In the DRIVE dataset, our TA-Mamba approach presents segmentation outcomes as follows: commencing with the original image and proceeding to the right, we display the ground truth, alongside the

segmentation results obtained from TA-Mamba, DSCNet, Att-UNet, UNet, and U-Net++

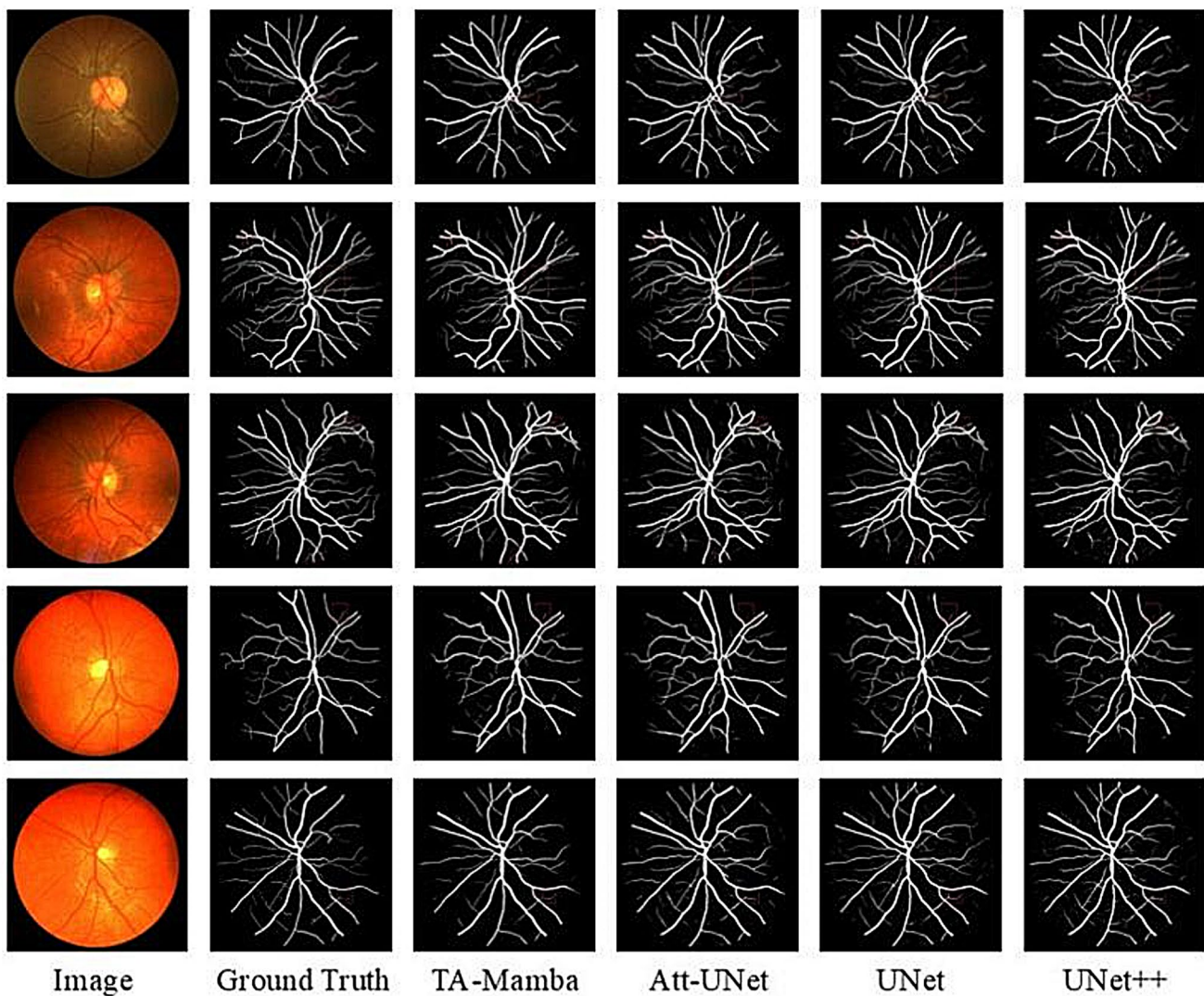


Fig. 7 In the CHASE_DB1 dataset, our TA-Mamba approach presents segmentation outcomes as follows: commencing with the original image and proceeding to the right, we display the ground truth, alongside the segmentation results obtained from TA-Mamba, Att-UNet, UNet, and U-Net++

Figures 6, 7, and 8 show the segmentation results on the three datasets, from which it can be seen that our method is more effective than others, but also performed poorly in some cases because the images themselves have extremely low contrast and severe noise.

4.2 Ablation study

To ascertain the efficacy of individual components, we conducted ablation studies on the STARE dataset. DSCNet [6] can be regarded as an encoder-decoder architecture composed of DSConvs, and we established five sets of experiments based on DSCNet as a benchmark to delineate the effectiveness of our components. The results of these experiments are presented in Table 4. Upon integrating serpentine SSC into the encoder of DSCNet, the model's performance

overall was enhanced, with improvements in Dice, cIDice, AUC, and IOU by 0.15%, 1.77%, 0.16%, and 0.21%, respectively. The experimental outcomes demonstrate that the SSC effectively assisted the model in focusing more on vessel structures. Subsequently, we incorporated the DFF module into the model, with the results as shown under “baseline + SSC + DFF” in Table 4. The extraction and fusion of directional features led to an increase in Dice, ACC, and IOU scores, substantiating the significance of directional features for retinal vessel segmentation. We then replaced the conventional upsampling with the CUB, and all evaluation metrics show an improvement over “baseline + SSC + DFF,” indicating that CUB aided the model in preserving rich information during the upsampling process. Finally, we further embed the HLMamba block into the model to validate the cumulative effect of all components, as illustrated under

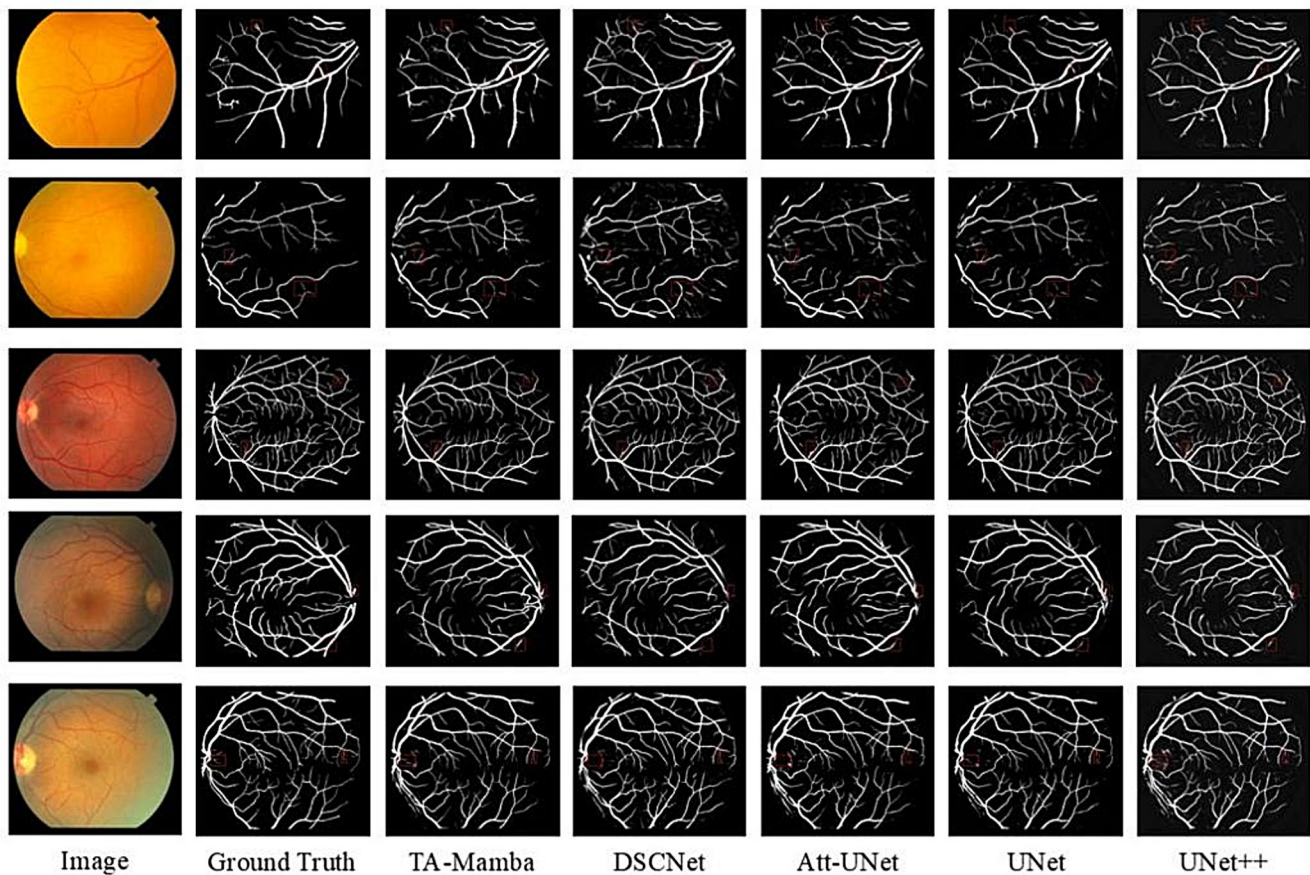


Fig. 8 In the STARE dataset, our TA-Mamba approach presents segmentation outcomes as follows: commencing with the original image and proceeding to the right, we display the ground truth, alongside the

segmentation results obtained from TA-Mamba, DSCNet, Att-UNet, UNet, and U-Net++

Table 4 Ablation study on the STARE dataset. Baseline: DSCNet, SSC: serpentine spatial convolution, DFF: directional feature fusion module, CUB: convolution-up block, HLMamba: high and low frequency attention Mamba block. Bolded data indicate the best. This kind of display continues for all subsequent experiments

Methods	Dice	clDice	ACC	AUC	IOU
Baseline	79.85	82.55	97.16	98.30	66.70
Baseline + SSC	80.00	84.32	97.13	98.46	66.91
Baseline + SSC + DFF	80.27	83.46	97.23	98.36	67.24
Baseline + SSC + DFF + CUB	80.95	85.01	97.31	98.57	68.19
Baseline + SSC + DFF + CUB + HLMamba	81.59	85.15	97.38	98.67	69.12

“baseline + SSC + DFF + CUB + HLMamba” in Table 4. This approach achieved the highest Dice, clDice, ACC, AUC, and IOU among the five experimental groups, thereby affirming the effectiveness of TA-Mamba in the task of retinal vessel segmentation.

4.3 Chunk size analysis

In this paper, we explored the effect of chunk size on the performance of TA-Mamba using the STARE dataset, and the specific results are shown in Table 5.

From Table 5, it can be seen that the segmentation performance of TA-Mamba is optimal when the patch size is set to 224. Therefore, we selected a patch size of 224 as the chunk size of the STARE dataset.

5 Conclusion

In this study, we focus on striking a balance between recognizing noise and preserving details. Firstly, we propose a high and low frequency attention Mamba block to capture global contextual information, thereby reducing the interference of noise. Secondly, we introduce tubular-aware gated convolution, serpentine spatial convolution and directional

Table 5 TA-Mamba performance analysis for different patch sizes on STARE dataset. Bolded data indicate the best. This kind of display continues for all subsequent experiments

Patch sizes	Dice	clDice	ACC	AUC	IOU	SE	SP
112 × 112	79.23	83.85	97.11	98.23	65.80	78.95	99.28
224 × 224	81.59	85.15	97.38	98.67	69.12	82.73	99.47
336 × 336	81.30	85.22	97.34	98.53	68.72	82.15	99.49

feature fusion module to accurately capture the features of tubular structures. Lastly, we propose a convolution-up block to reduce the loss of information during the upsampling process. Our method is verified on DRIVE, CHASE_DB1, and STARE datasets and the results show our method outperforms state-of-the-art methods in terms of accuracy, continuity, and topological connectivity. It achieves a good balance between discriminating noise and preserving details, effectively addressing the challenges posed by low contrast and high noise in retinal images.

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Author contributions DangGuo Shao: Conceptualization, Writing - Review & Editing, Supervision, Resources, manuscript revision. Rui Xu: Conceptualization, Methodology, Code, Writing - Original Draft, Writing - Review & Editing, Visualization, manuscript revision. Lei Ma: Conceptualization, Supervision, Resources, manuscript revision. Sanli Yi: Supervision, Resources, manuscript revision.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Declaration of originality We would like to state that this work is original research that has not been published before and is not considered for publication elsewhere.

Ethics approval and consent to participate This article does not contain any studies with human participants performed by any of the authors.

Competing interests The authors declare no competing interests.

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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